

Zhihao Wang

Postdoctoral Researcher, University of Maryland

World Models | Scientific Machine Learning | Geospatial Computer Vision

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Education

University of Maryland, College Park	Ph.D., Geographical Sciences	2020–2026
The Ohio State University	M.A., Geography	2018–2020
University of Waterloo	B.E.S., Honours Geomatics; Minor in Computer Science	2016–2018
Wuhan University	B.E., Remote Sensing Science and Technology	2014–2018

Research Experience

First-author publications: NeurIPS (2, Datasets and Benchmarks), AAAI (1), and ACM SIGSPATIAL (2).

University of Maryland, College Park 2020–Present

Postdoctoral Researcher (Jun 2026–Present); Doctoral Researcher (2020–2026)

World Models & Scientific Machine Learning SIGSPATIAL 2026

- Developed an **action-conditioned Earth-system world model** using transition-action pretraining and masked response learning, enabling user-specified state edits without manually annotated intervention labels; evaluated coupled ecosystem responses across **six global regions**.
- Built **Deep-ED**, a deep learning emulator of the EDv3.0 ecosystem simulator; combined multi-scale temporal modeling, de-sequencing, and knowledge-guided partitioning to reduce error accumulation in long-horizon forecasts.
- Created and released **CarbonGlobe**, a global **40-year benchmark at 0.5° resolution**; evaluated Crossformer, DLinear, and TimeXer under climate-driven distribution shift. *Code | Data*

Foundation Model Evaluation & Generalization 2024–2026

- Led evaluation of **10 geospatial and vision foundation models across 10 satellite benchmarks**, including AlphaEarth, Prithvi2, DINOv3, and SpectralGPT; compared task- and resolution-dependent generalization with DeepLabV3+ and SegFormer.
- Developed an evaluation framework linking spatial errors to confidence intervals and field-validation costs, explaining why models with similar F1 scores can require different sampling effort.

Computer Vision & Multimodal Model Routing NeurIPS 2025; ICDMW 2025

- Developed **TreeFinder**, an open-source tree-mortality segmentation benchmark with pixel-level annotations spanning **48 U.S. states and 23,000+ hectares** of 0.6 m aerial imagery; designed evaluation across geographic regions, climates, and forest types. *Code | Data*
- Co-developed **MARS**, using multimodal foundation models to route aerial images to scenario-specialized segmentation models without changing backbone architectures.
- Produced 10-year deforestation maps from approximately **15 TB of satellite imagery at 4.7 m resolution** to assess forest loss and limitations in existing maps.

Physics-Guided Learning for Geographic Fairness AAAI 2024

- Developed **SimFair**, reducing geographic error disparity by **73% on average across nine spatial splits** versus a feedforward baseline, without target-region training labels; used an invertible network to align learning with a physical simulator.

National Oceanic and Atmospheric Administration (NOAA) May 2021–Aug 2022

Graduate Research Assistant

- Developed long-term albedo climatology for the **NOAA-20 VIIRS Global Albedo product**, processing **3.5+ TB of satellite observations** with parallel workflows on a Linux cluster.
- Designed memory-efficient statistical processing for large satellite datasets and contributed to testing operational satellite algorithms.

Selected Publications

First-author papers Full publication list: Google Scholar

- **Wang, Z.**, et al. Earth System World Model for What-If Simulations: A Case Study for Terrestrial Ecosystems. *ACM SIGSPATIAL 2026*.
- **Wang, Z.**, et al. CarbonGlobe: A Global-Scale, Multi-Decade Dataset and Benchmark for Carbon Forecasting in Forest Ecosystems. *NeurIPS 2025, Datasets and Benchmarks Track*.
- **Wang, Z.**, et al. TreeFinder: A US-Scale Benchmark Dataset for Individual Tree Mortality Monitoring Using High-Resolution Aerial Imagery. *NeurIPS 2025, Datasets and Benchmarks Track*.
- **Wang, Z.**, et al. SimFair: Physics-Guided Fairness-Aware Learning with Simulation Models. *AAAI 2024*.
- **Wang, Z.**, et al. High-Fidelity Deep Approximation of Ecosystem Simulation over Long-Term at Large Scale. *ACM SIGSPATIAL 2023; Oral Presentation*.

Selected collaborative papers

- Li, R., **Wang, Z.**, et al. EcoDiffusion: Uncertainty-Aware Emulation of Ecosystem Processes with Conditional Diffusion for Long Sequences with Single-Step Initialization. *AAAI 2026*.
- Li, C., **Wang, Z.**, & Xie, Y. Multimodal Foundation Models as Router Models for High-Resolution Aerial Image Segmentation. *IEEE ICDM Workshops 2025*.

Preprints & Manuscripts

- **Wang, Z.**, et al. Re-Imagining Foundation Models for Earth Monitoring: From Larger Models and Benchmark Gains to Operational Value. *Manuscript in preparation, 2026*.
- Mai, G., Rao, J., Cao, Q., Liu, Z., **Wang, Z.**, et al. Towards Generative Interactive Geospatial World Models for Proactive and Scenario-Driven Geospatial Decision-Making. *Manuscript under review, 2026*.
- Xie, Y., **Wang, Z.**, et al. When are Foundation Models Effective? Understanding the Suitability for Pixel-Level Classification Using Multispectral Imagery. *arXiv:2404.11797, 2024*.

Technical Skills

Programming: Python, C/C++, SQL, MATLAB, R, JavaScript.

ML frameworks & methods: PyTorch, TensorFlow; world models, physics-guided learning, time-series forecasting, semantic segmentation, foundation-model evaluation, geographic generalization.

Data & research tools: Linux, parallel data processing, Git, Google Earth Engine, Google Cloud, ArcGIS.

Selected Honors

First Place (tie) , UCGIS Student Lightning Talk Award	2026
Outstanding Graduate Assistant Award , University of Maryland	2026
Scholar Award (travel) , NeurIPS	2025
Student Scholarship (travel) , AAAI	2024
Excellence in Graduate Research Award (Second Place) , University of Maryland Department of Geographical Sciences	2024
Travel Award , ACM SIGSPATIAL GIS Cup	2023
Third Place , ACM SIGSPATIAL GIS Cup	2022

Research Leadership & Service

- Organizing committee: ACM SIGSPATIAL GeoIndustry workshop (2025, 2026).
- Program committee/reviewer: NeurIPS 2026, KDD 2025–2026, AAAI 2026, ACM SIGSPATIAL 2025–2026.
- Mentored three student researchers (2024–2025). Instructor for Python Programming and Geocomputing, University of Maryland (2025).